

P2

Infographics & Diagrams

Visual Data, Frameworks & Sector Snapshots

PART 2 — INFOGRAPHICS & DIAGRAMS

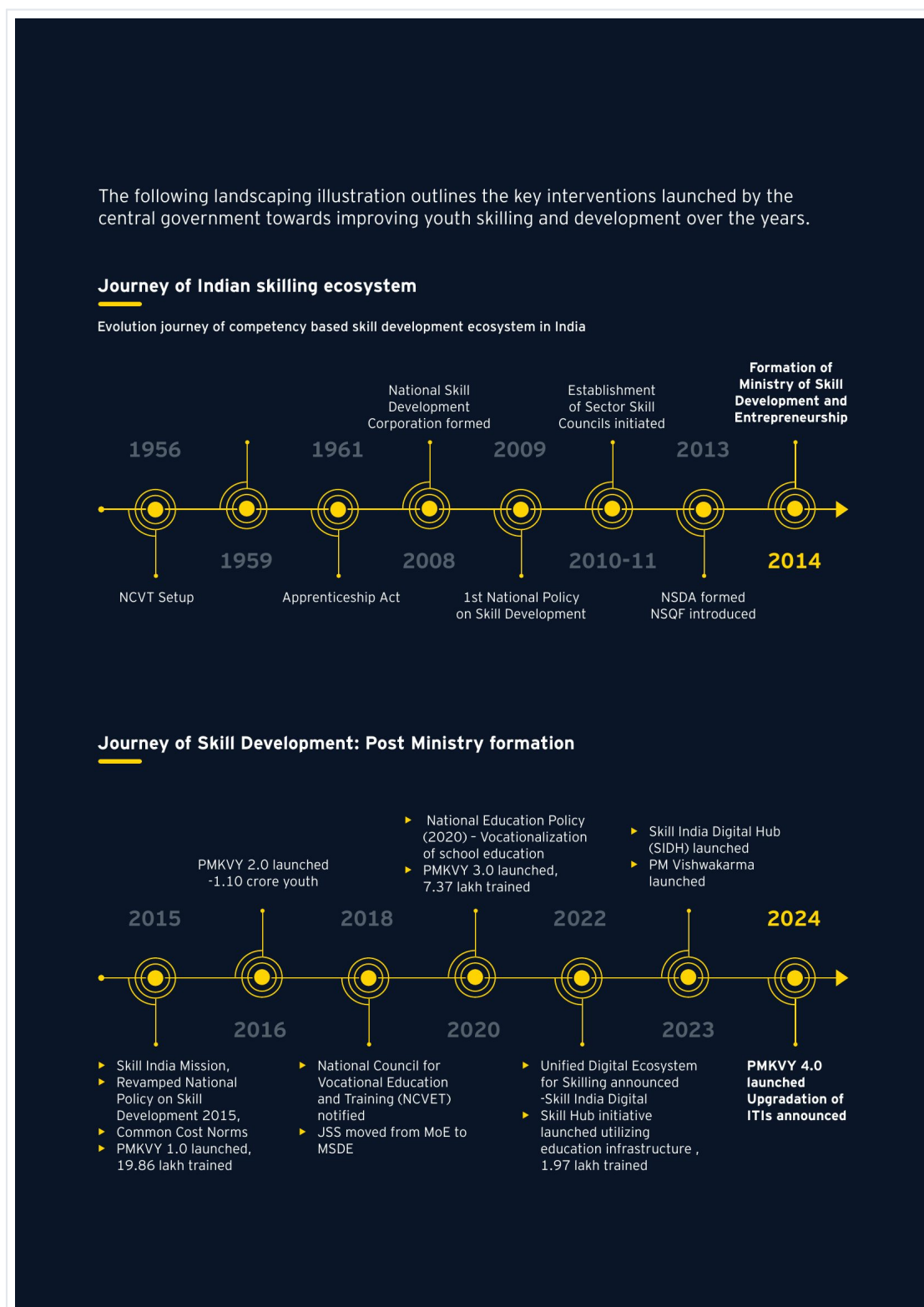
Visual Index

This section brings together the report's key diagrams, frameworks and data visualisations in one place — the skilling-ecosystem timeline, the study's methodology framework, sector-wise survey charts, and the closing stakeholder-readiness matrix. Each figure is reproduced directly from the source report and annotated with a short caption for quick reference.

FRAMEWORKS	SECTOR CHARTS	SURVEY DATA	READINESS MATRIX
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FRAMEWORK · POLICY TIMELINE

Figure 1 — Journey of the Indian Skilling Ecosystem (1956–2024)



A timeline diagram tracing the evolution of India's competency-based skill development system — from the founding of the NCVT in 1956 through PMKVY 1.0–4.0, the National Education Policy 2020, and the 2024 launch of the Skill India Digital Hub and PM Vishwakarma scheme.

FRAMEWORK · RESEARCH DESIGN

Figure 2 — FOJ 3.0 Study Methodology — 5 Sectors × 5 Lenses × 3 Personas

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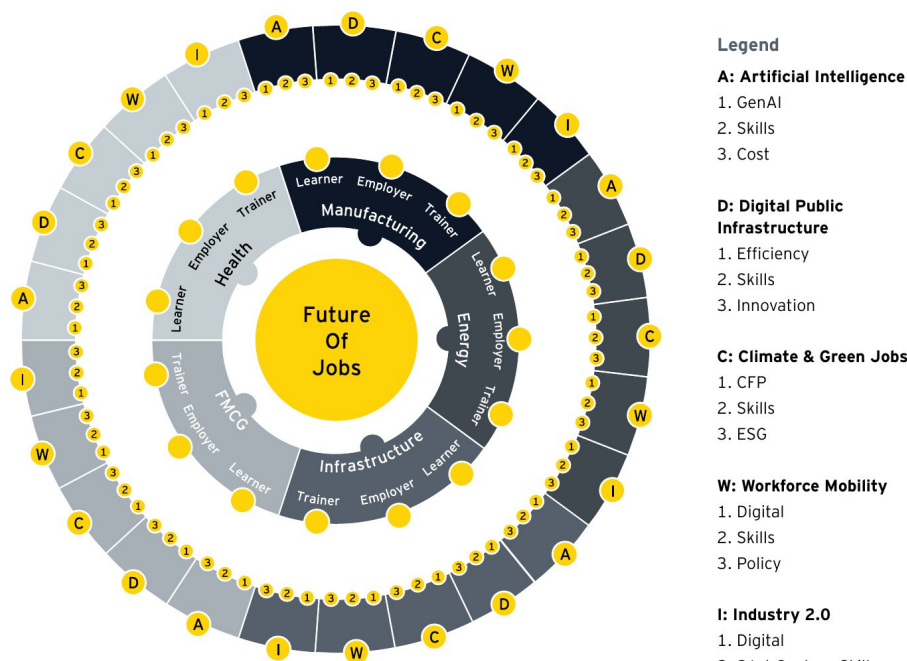
FICCI-EY future of jobs reports

As the job and skilling landscape evolves at a rapid pace with disruptive technologies making way, and COVID-19 pandemic which brought significant changes to the labor market, a committee comprising senior industry experts and panel of FICCI and EY made an informed decision to publish a new edition of FOJ report which studies in-depth the jobs and skills landscape of India across five sectors, namely manufacturing, FMCG, healthcare, energy and infrastructure.

Study methodology

For this report, the committee outlined a framework focused on assessing the impact of five trends– Artificial Intelligence (AI), DPI, Climate, Workforce Mobility and Industry 4.0– on the five sectors and how its interplay will affect three key stakeholders– learners, employers and trainers in the future in terms of jobs and skills.

We further sub-divided the five lenses into key factors, which we then individually considered to assess the impact on all sectors and stakeholders. We completed the assessment with the help of primary (interviews and digital survey) and secondary research (literature review) and then analyzed and drew insights from the data collected.



Sectors (5): Manufacturing, Health, FMCG, Infrastructure, Energy

Lens (5): Artificial Intelligence (AI), DPI, Climate, Workforce Mobility and Industry 4.0

Personas/Stakeholders (3): Employer (Industry), Learner (in education, skilling-upskilling and re-skilling) and Trainer (Industry, private sector and Government)

In the concluding section, we have presented our findings of the assessment in the form of Harvey Balls visualization.

The concentric framework diagram used to structure the entire study: five sectors (Manufacturing, Health, FMCG, Infrastructure, Energy) are each assessed through five lenses (AI, DPI, Climate, Workforce Mobility, Industry 4.0) across three stakeholder personas (Employer, Learner, Trainer).

SECTOR CHART · MANUFACTURING

Figure 3 — Manufacturing — AI Adoption Readiness & Government Incentives

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3.1.4

Future of jobs: insights from industry leaders

The primary survey has reached out to industries with strength of more than 500 employees working with their presence in regional, national and international markets.

Emerging job roles in the sector

The future of jobs in the manufacturing sector is poised to be shaped by roles that capitalize on advanced technologies. According to the survey, 60% of respondents anticipate significant growth in jobs for AI and robotics technicians, as well as IoT specialists, 40% foresee high demand for sustainable manufacturing engineers and digital supply chain analysts.

The survey further reveals that 40% of respondents believe skills in robotic programming, advanced analytics, and Internet-of-Things connectivity will be most in demand for future manufacturing roles. Only 20% expect a high demand for AI and machine learning application skills, which presents an interesting contrast, highlighting the evolving landscape of skill requirements in the sector.

Impact of GenAI, AI, sector-specific technology, international mobility, DPI, climate and green jobs on the job and skilling ecosystem

The impact generated by generative AI and AI technologies in the manufacturing sector is likely to be very strong. The use of these technologies may likely result in creating new job roles and altering the nature of existing ones, particularly in the case of supply chain

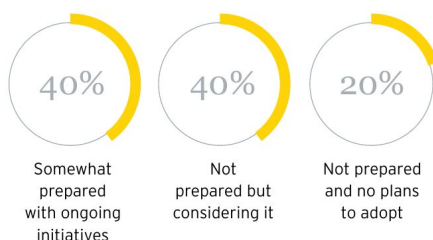
optimization and predictive maintenance. The survey revealed concerns related to sector preparedness in picking up these technologies. As much as 40% of the respondents registered ongoing initiatives in AI adoption preparedness, while 40% mentioned that organizations were not prepared at all and only thinking about it. By itself, this presents an enormous gap between recognition of AI potential and readiness towards its realization. Secondly, majority respondents highlighted the major challenges in AI integration are workforce's resistance, and lack of technical skills. The other factor identified was data privacy and security concerns. This is also related to secondary research, emphasizing a need for upskilling and reskilling initiative to face these challenges. The gap that ultimately exists between the projected level of influence that AI will have and the levels of preparedness and usage actually being at work within the manufacturing sector speaks to the fact that a more concerted approach needs to be taken with regards to the way in which such technologies will be integrated.

The surveyed organizations are increasingly adopting sustainability practices, with organizations implementing programs such as energy-efficient machinery (50%), waste reduction (33%), and sustainable sourcing (33%). AI and generative AI are expected to significantly impact supply chain optimization (44%), predictive maintenance (36%), and production line automation (27%).

Suggested programmatic and policy interventions

Most of the industry representatives felt that the program and policy interventions had to match the dynamism of the sector. 80% of the respondents felt that governmental support in creating renewable energy infrastructure would be beneficial, 40% mentioned the importance of stringent environmental regulations to encourage sustainability. Equally, 40% believe that the grants for acquiring new equipment and software would be the most effective government incentives towards technological advancement. Out of the total workforce sample, 80% preferred a supportive government partnership with the private while none suggested the government to be at the frontline in training and education. This testifies to the preference for public-private collaboration over government initiatives in a situation where the urgency for skill development is well recognized.

Preparedness of AI adoption



Survey results on organisational preparedness for AI adoption in manufacturing, alongside the government incentives industry leaders find most beneficial for technological advancement.

SECTOR CHART · MANUFACTURING

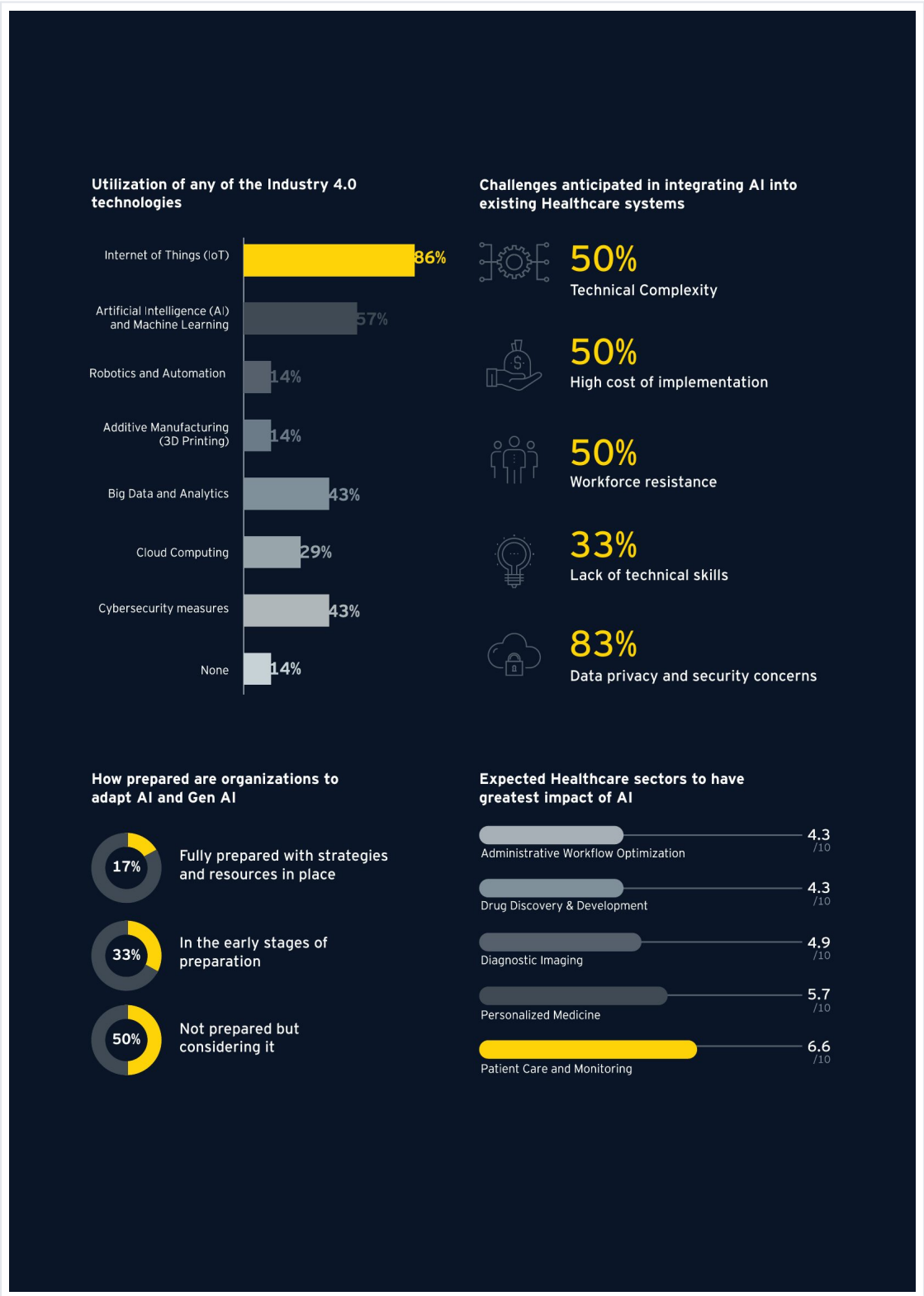
Figure 4 — Manufacturing — Emerging Job Roles & In-Demand Skill Sets



Bar charts ranking the manufacturing job roles expected to emerge (AI & Robotics Technician, IIoT Architect, Sustainable Manufacturing Engineer) against the skill sets employers say will be most valuable — robotics programming, advanced analytics, IoT connectivity and digital literacy.

SECTOR CHART · HEALTHCARE

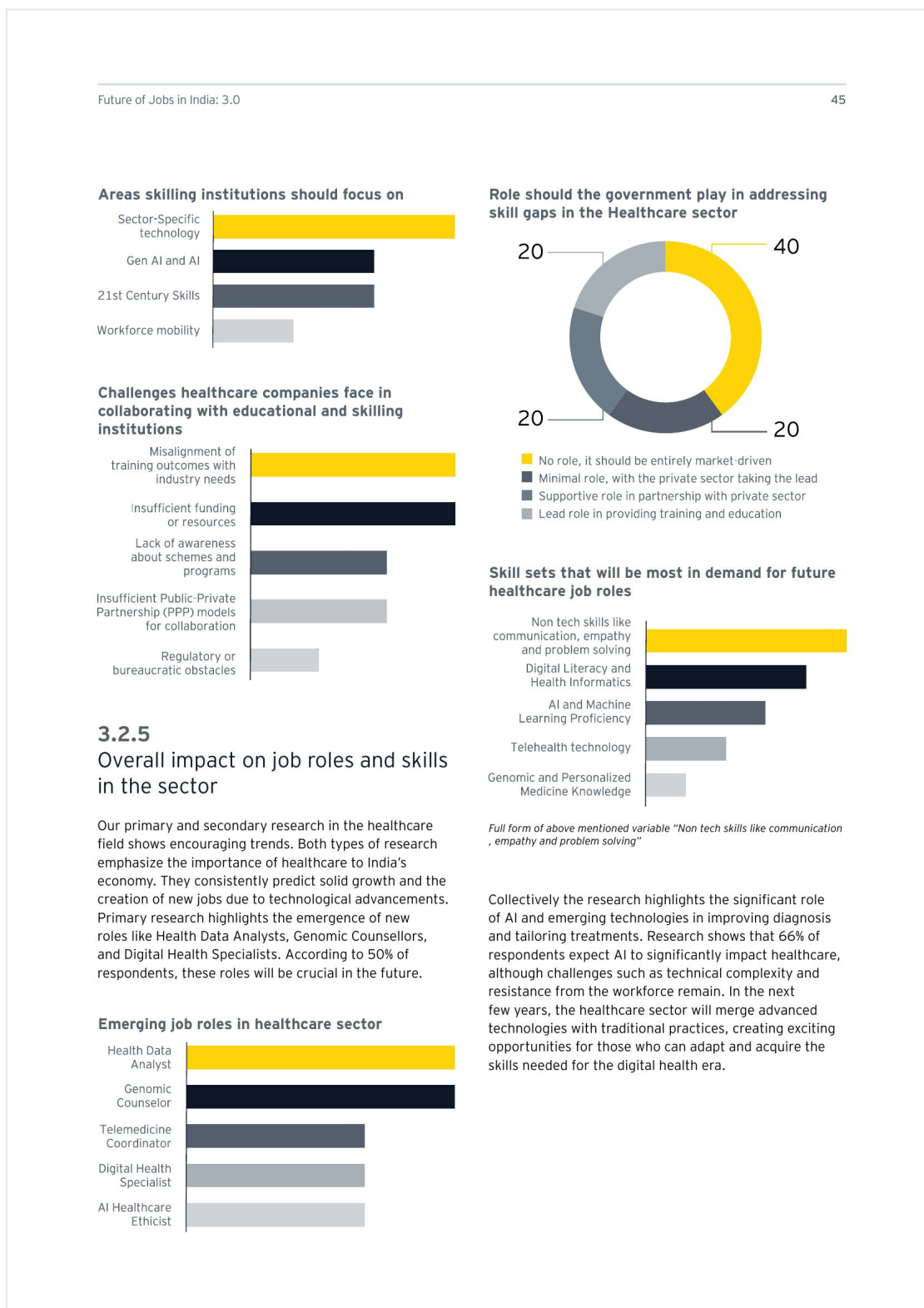
Figure 5 — Healthcare — Industry 4.0 Utilisation & AI Integration Challenges



Adoption levels of IoT, AI/ML, Big Data and cybersecurity tools in Indian healthcare, paired with the leading challenges providers anticipate when integrating AI into clinical systems.

SECTOR CHART · HEALTHCARE

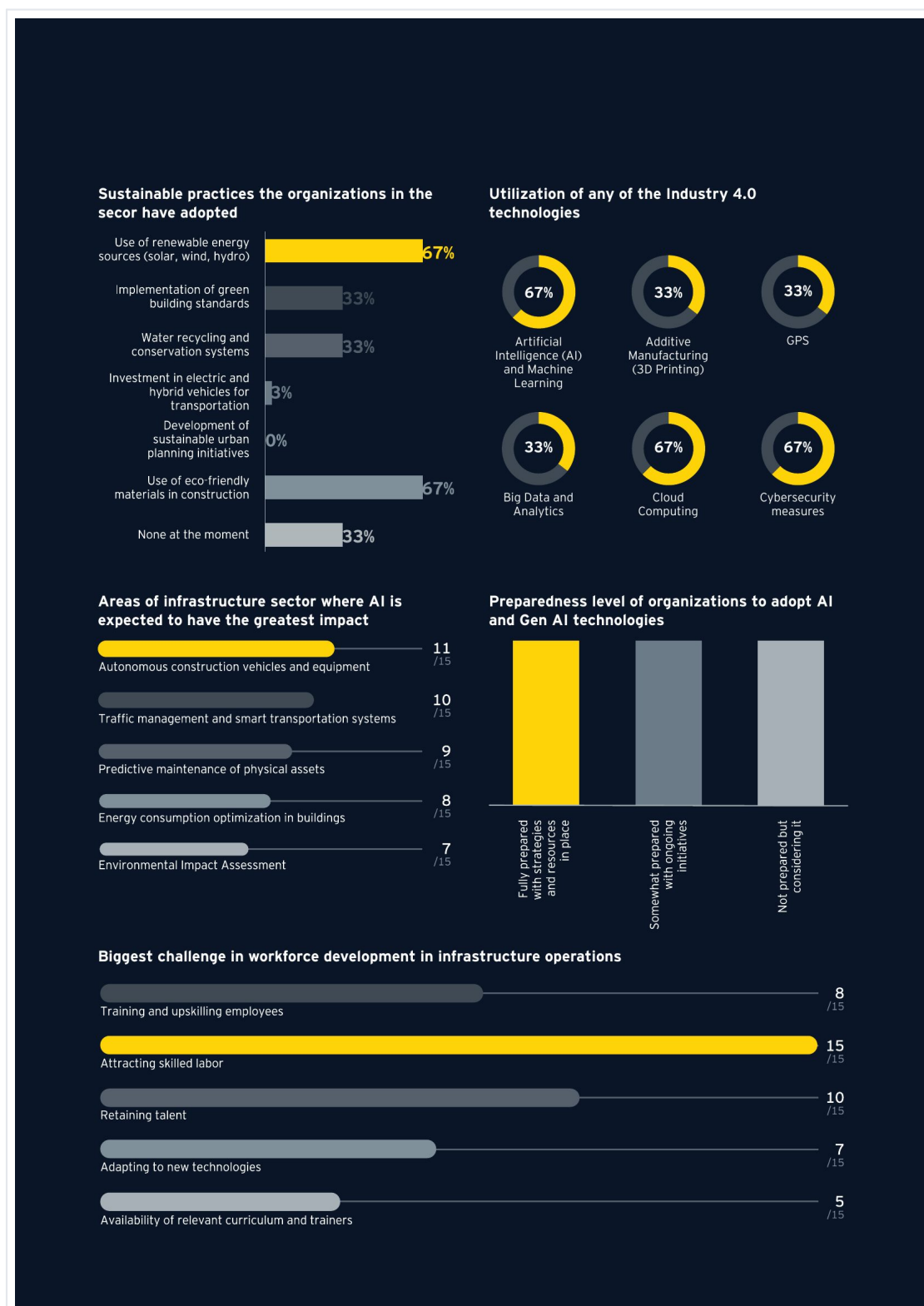
Figure 6 — Healthcare — Emerging Job Roles & Skilling Priorities



Health Data Analyst, Genomic Counsellor, Telemedicine Coordinator and Digital Health Specialist emerge as priority roles, with non-technical (empathy, communication) and digital-health literacy skills ranked as most critical for the sector's future workforce.

SECTOR CHART · INFRASTRUCTURE

Figure 7 — Infrastructure — Sustainable Practices & Industry 4.0 Adoption



The sustainability measures infrastructure organisations have already adopted (renewable energy, eco-friendly construction materials) alongside their current use of Industry 4.0 technologies.

SECTOR CHART · INFRASTRUCTURE

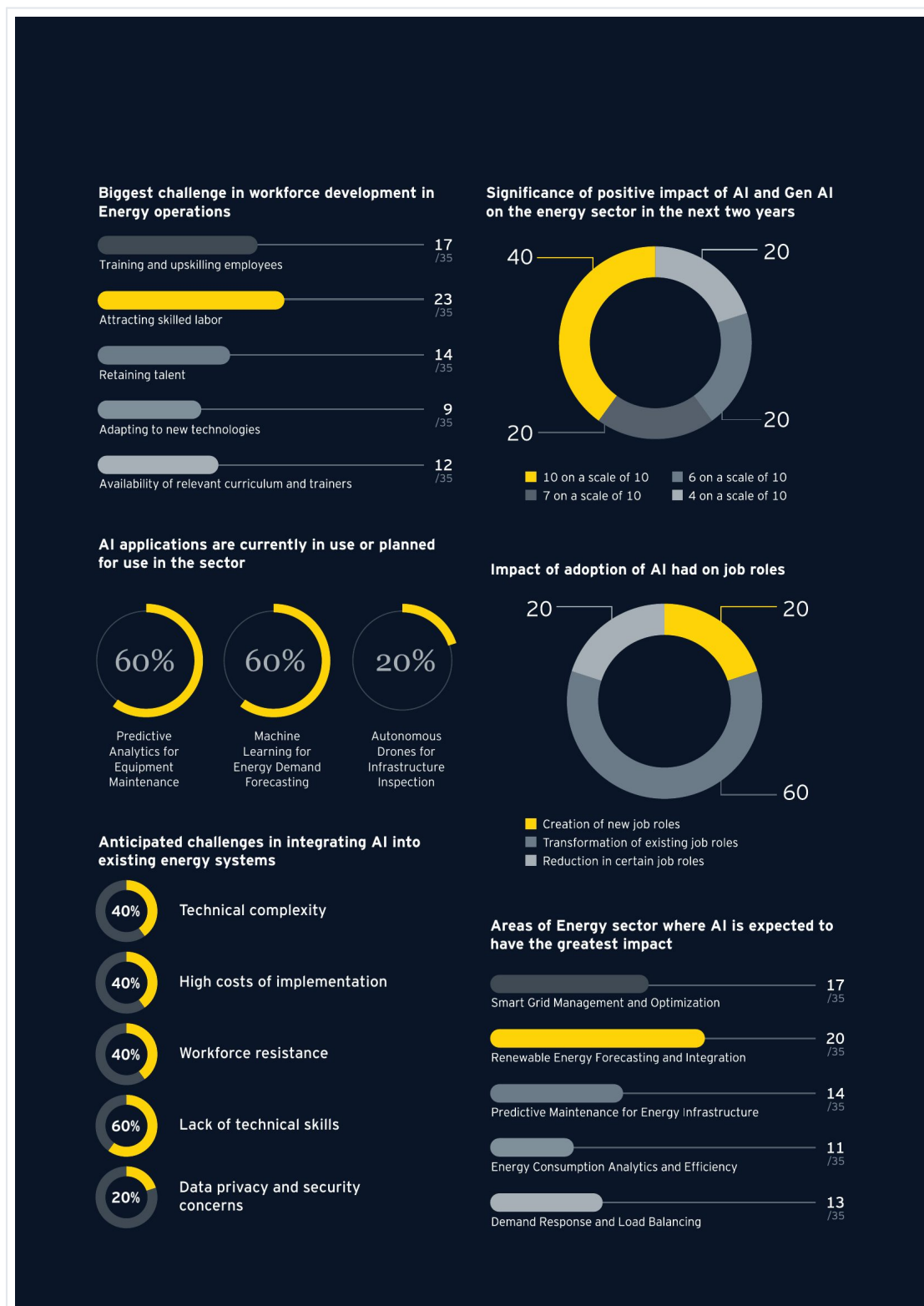
Figure 8 — Infrastructure — Emerging Roles & Required Expertise



AI & Machine Learning Specialists, Renewable Energy Technicians and Smart City Planners lead the infrastructure sector's emerging-roles list, with sustainable design and green-technology expertise topping the required skill sets.

SECTOR CHART · ENERGY

Figure 9 — Energy — Workforce Development Challenges & AI Readiness



The biggest workforce-development hurdles in energy operations (attracting skilled labour tops the list) alongside the sector's expected timeline for realising positive AI and Gen-AI impact.

SECTOR CHART · ENERGY

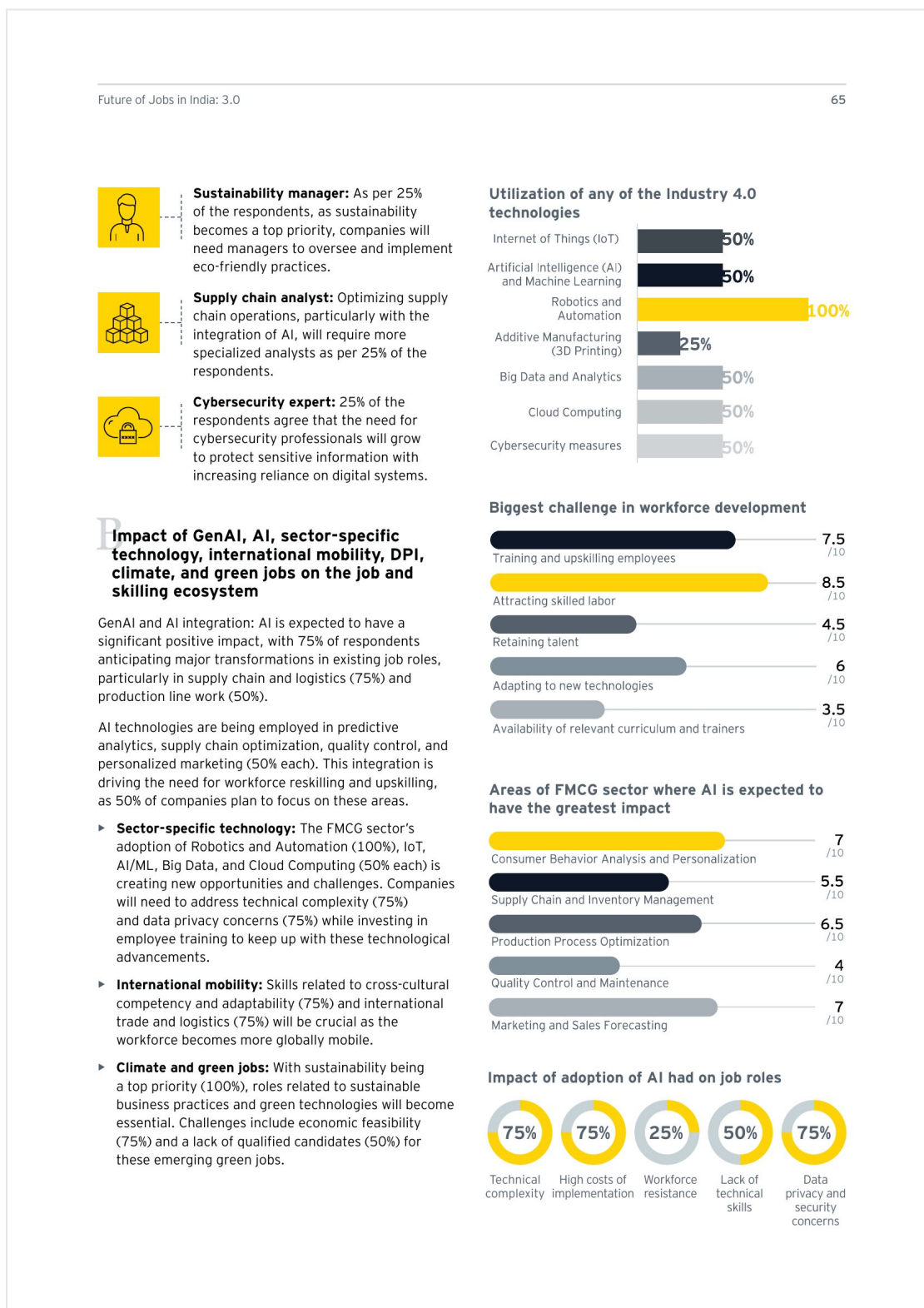
Figure 10 — Energy — AI Impact Areas, Job Transformation & Emerging Roles



Smart-grid management and renewable-energy forecasting are rated the areas most likely to feel AI's impact; most respondents expect AI to transform — rather than eliminate — existing energy-sector roles.

SECTOR CHART · FMCG

Figure 11 — FMCG — Industry 4.0 Utilisation & AI Integration Outlook



Robotics and automation show the highest adoption in FMCG operations, while the sector anticipates AI will chiefly transform existing job roles in supply chain, logistics and production-line work.

SECTOR CHART · FMCG

Figure 12 — FMCG — Emerging Job Roles & Future Skill Sets



Data Scientist and AI & Robotics Technician rank as the top emerging FMCG roles, with sustainable business practices and green technologies identified as the leading future skill requirement.

CROSS-SECTOR SNAPSHOT

Figure 13 — Cross-Sector Snapshot — Emerging Job Roles (All 5 Sectors)

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4.2.9

AI- Assistant

AI as a Knowledge Assistant can help targeted beneficiaries such as residents, government officers, ecosystem partners, the authority and other relevant stakeholders to access, just in time, context relevant information from trusted sources.

Benefits:

- ▶ **Efficiency:** Enhance efficiency in information dissemination and subsequently in processes, ensure accuracy and optimization of processes, build trust, and improve user experience and satisfaction
- ▶ **Speed:** Increased and accelerated enrolments/onboarding and adoption- enrolling institutions seamlessly without unnecessary and avoidable bottlenecks
- ▶ **Single interface:** Reduce turnaround time for issue addressal and unified interface for multiple sources and for multiple users



4.3

Snapshot- Future of Jobs 3.0

This section summarises primary research findings on sectoral job roles and skills. The detailed findings have been shared in the previous section of the report.

● Emerging job roles in all 5 sectors ●

Manufacturing	Healthcare	Infrastructure	Energy	FMCG
- AI & Robotics technician	- Health Data Analyst	- AI and Machine Learning Specialists	- Renewable Energy Technicians	- Data Scientist
- Industrial Internet of Things (IIoT)	- Genomic Counselor	- Renewable Energy Technicians	- Smart Grid Analyst	- AI and Robotics Technician
- Sustainable Manufacturing Engineer	- Telemedicine Coordinator	- Sustainable Infrastructure Engineers	- Sustainability Managers	- Digital Marketing Specialist
- Digital Supply Chain Analyst	- Digital Health Specialist	- Smart City Planners	- Carbon Capture Technician	- Sustainability Manager
- Data Privacy and Cyber Security Manager	- AI healthcare Ethicist	- Warehousing operators & technicians	- Audit Manager	- E-commerce Strategist
- Maintenance Technicians	- Caregivers	- CNC machine operators	- Environment health safety specialist	- Computer operators
- Design Engineers	- Geriatric Caregivers	- Research & Development (R&D)	- Supply chain Analyst	- Supply Chain Analyst
- Welders	- Mental wellness experts	- Ground Liaison Officers	- Social Impact Manager	- Data Visualizer
- Digital Construction Engineers	- Energy Conservation-alists		- Compliance Manager	- Data Engineers
	- Remote Patient Monitoring Specialists		- Green Supply Managers	- Omni Channel Specialist
			- Sensitivity Analyst	

A consolidated view of emerging job roles across Manufacturing, Healthcare, Infrastructure, Energy and FMCG, drawn directly from the primary survey of industry leaders in each sector.

CROSS-SECTOR SNAPSHOT

Figure 14 — Cross-Sector Snapshot — Top Skills & Stakeholder Preferences

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● Top skills in all 5 sectors ●

**In addition to the skills required to meet the job roles listed above, the primary findings also list the below mentioned skills:*

Manufacturing	Healthcare	Infrastructure	Energy	FMCG
- Robotics programming and maintenance - Advanced analytics and data interpretation - Internet of Things (IoT) and connectivity - Advanced digital literacy - Soft skills (e.g., problem-solving, communication)	- Non tech skills like communication, empathy and problem solving - Digital Literacy and Health Informatics - AI and Machine Learning Proficiency - Telehealth technology - Genomic and Personalized Medicine Knowledge	- Expertise in sustainable design and green technologies - Understanding of advanced materials and construction techniques - Robotics and Automation	- Expertise in Renewable Energy Technologies - Skills in Energy Data Analytics and AI - Proficiency in Energy Storage Solutions - Knowledge of Environmental and Energy Policy	- Sustainable Business Practices and Green Technologies - Consumer Behavior Research and Insights - Data Analysis and Interpretation - Digital Literacy and E-commerce Management - Supply Chain and Logistics Optimization

Top skill sets by sector, alongside learner- and trainer-reported views on preferred skilling mechanisms, high-growth sectors and the technologies expected to reshape jobs and skills.

READINESS MATRIX

Figure 15 — Readiness Matrix — Impact of AI, DPI, Climate, Mobility & Industry 4.0

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4.4

Are you ready- Learner, Trainer, Employer

With reference to the framework mentioned in section 2.3, presented here is a summary of the results from our primary and secondary research highlighting the impact of the five lenses used in this report– Artificial Intelligence, DPI, climate, international mobility and Industry 4.0, on the sectors manufacturing, health, infrastructure, energy and FMCG vis-a-vis employer (industry), learner (in education, skilling-upskilling and re-skilling) and trainer (industry, private sector and government).

Key Factors

Artificial Intelligence

- Generative AI
- Skills required for adoption of AI
- Cost of implementation

DPI

- Efficiency of sector specific DPI
- Skills required for implementation of DPI
- Sector specific Innovation

Climate

- Carbon Foot Print
- Skills required for implementation of climate specific requirements
- Environmental, social and governance

International Mobility

- Adoption of Digital platforms
- Sector Specific Skills
- Policies for Governance, Compliance, Investment etc

Industry 4.0

- Adoption of Digital platform
- 21st century skills
- Policies for Governance, Compliance, Investment etc

Scale

- No Impact
- Low Impact
- Moderate Impact
- High Impact
- Very High Impact

Following analysis looks at the impact of key impact areas in each lens from maximum impact (1=100%) followed by 25% reduction in each. The below framework helps to assess how each trend will impact Learners, Trainer and Employers.

Sector	Eco Play	Artificial Intelligence			DPI			Climate			Workforce Mobility			Industry 4.0		
		GenAI	Skills	Cost	Efficiency	Skills	Innovation	CFP	Skills	ESG	Digital	Skills	Policy	Digital	21st Century Skills	Policy
Manufacturing	Employer Employer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Learner	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Trainer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Health	Employer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Learner	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Trainer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Infrastructure	Employer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Learner	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Trainer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Energy	Employer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Learner	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Trainer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
FMCG	Employer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Learner	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Trainer	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

The report's signature Harvey-ball matrix, scoring the impact of all five lenses on Learners, Trainers and Employers within each of the five sectors — from no impact to very high impact.